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Cloud Computing
CSA1514

ASSESSMENT TOOL: 1

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1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

cloud computing is the delivery of shared computing resources, such as servers, storage, databases, and applications, over a network, usually the internet. Users can obtain and release resources on demand with minimal provider interaction.

characteristic	cloud computing	Traditional on-premise Computing.
* On-demand self service	Users provision resources whenever needed without manually contacting the provider.	IT states usually purchase, install, and configure resources.

characteristic	cloud Computing	Traditional on-premise computing.
Broad Network access	Services are accessible over a network thrugh Computers, phones, and other devices.	Access depends on the organisation's network and remote-access arrangements.
Resource pooling	Providers share infrastcre among multiple customers while logically isolates their resources.	Infrastucture is generally dedicated to one organisation.
Rapid elasticity	Resources can expand or shrink quickly according to demand.	Expansion usually requires purchasing and installing additional hardware

Example :- An Online Shopping website can temporiarily increases its cloud Server capacity during a festival Sale and reduce it afterward.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution and show how these changes enabled modern cloud platforms.

Cloud computing developed through improvements in both hardware and internet software.

1. Mainframes and time-sharing : Many users accessed one powerful and computer through terminals. This introduced the idea of sharing computing resources.

2. Personal Computers and client-server Systems :- Affordable microprocessors enabled personal computers. client-server networks allowed users to access centrally managed applications and data.

3. Virtualisation :- A hypervisor enabled multiple virtual machines to run on one physical server. This improved hardware utilisation and allowed workloads to remain isolated.

4- Distributed computing and data centres :- Networked servers worked together to process tasks and store data. Large data centres provided computing capacity at scale.

5. Internet and web software :- Wider internet access, web browsers, web services, and APIs enabled applications to be accessed remotely and systems to communicate.

6. Modern cloud platforms :- Provides combined virtualization, large data centres, and automated resources management to deliver computing servies on demand.

Therefore, hardware evolution supplied scalable computing capacity, while internet software made that capacity remotely accessible and programmable.

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8. Explain the role of server virtualisation in cloud computing. Differentiate between virtualisation as an enabling technology and cloud computing as a Service model.

Sever Virtualization divides a physical server into multiple Virtual machines (VMs) using software called a hypervisor. Each VM has Virtual CPU, Memory, Storage and its own operating System.

Roles in cloud Computing :-

* Improves hardware utilisation by sharing resources.
* Isolates workloads in separate VMs.
* Enables quick creation and flexible allocation of resources.
* Supports backup, migration, and recovery.

Virtualisation	cloud Computing
Technology for creating virtual resources.	Models for delivering computing services over a network.
Focuses on hardware sharing and isolation	Provides on-demand access, scalability and measured usage.
Example: VMware, Esxi	Example: Amazon EC2.

4. Discuss the role of data centres in delivering clou Services.

A data centre is a facility containing servers, Stora systems, networking equipments, and supporting infrastructure It provides the physical foundation for cloud services.

Role	Explanation
★ Computing and Storage	Servers execute applications, while Storage systems hold files, databases, and backups.
★ Network connectivety	Switches, routers, and network links connect resources and deliver services to users.
★ Resource sharing	Virtualisation and management software divide physical infrastructure into resources for different customers.
★ Availability	Redundant power, network connections, and hardware reduce disruption caused by individual component failures.
★ Security	physical access controls, surveillance, and network security help protect infrastsure and data.

Example: A cloud-hosted application can run across multiple availability zones. If one zones becomes unavailable, a correctly configured application can continue operating from another zone.

Thus, data centres provide the capacity, connectivity, protection, and reliability needed to deliver cloud services.

5. Differentiate Iaas, Paas, Saas, and xaas with one suitable real-world example for each model.

Cloud services models differ mainly in what the provider delivers and what the customer manages.

Models	Full Form	Provides	Example.
Iaas	Infrasture as a services	Virtual servers, storage, and network users manages the OS and application	Amazon EC2
Paas	Platform as a Services	A managed platform where users deploy application code	Google App Engine.
Saas	Software as a services	Ready-to-use software accessed Online	Gmail
Xaas	Anything as a Services	An umbrella term covering IT Services	Amazon RDS

Main difference:

Iaas provides infrastructure, Paas provides an application platform, and Saas provides finished software xaas is the broader category that includes these and other services offerings.

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